

Effect of Positive Productivity Shock in the Energy Industry on Different Economic Variables

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Introduction

Energy-intensive regions such as Wyoming have significant energy and growth challenges in the 21st century with the ever-increasing demand for energy. To address these challenges, it is crucial to implement appropriate policies that aim to optimize social welfare not just inside the region itself but also in the areas where energy is exported. This is particularly important for regions that suffer from energy deprivation, as they can greatly benefit from the efficient utilization and distribution of energy resources. The formulation of policies should aim to mitigate the adverse effects on other industries resulting from the rapid growth in a particular sector while also striving to minimize the overall costs associated with implementing such policies. This aspect holds significant importance due to the fact that energy pricing and supplies garner a disproportionate level of attention from the media, legislators, voters, and economists alike. This disparity arises due to the distinct behavior exhibited by energy price and productivity shocks in comparison to other economic shocks. Typically, energy shocks have a pronounced and prolonged nature, hence exerting a substantial influence on the decision-making of both individuals and businesses. Given the inelastic nature of energy consumption, these shocks have a considerable impact on the decisions made by these economic agents. Throughout history, there has been a correlation between energy shocks and subsequent shocks in significant economic indicators such as GDP growth, inflation, and unemployment. Therefore, it is imperative to do a comprehensive analysis of the ramifications of an energy shock within an economy that encompasses numerous industries with differing levels of factor and trade intensity. This research adopts a simplified approach by considering only labor and capital as the primary elements for four industries (or goods) in Wyoming, including agriculture, energy, recreation, and miscellaneous. The latter sections of the article are dedicated to conducting an analysis of the impact of a positive shock within the energy industry, specifically in the form of increased factor productivity. This analysis encompasses 16 economic variables, namely compensating variation (CV), price of domestically produced goods (PD), wage rate (w), rental rate (r), price of total domestic and foreign consumption (PX), price of total domestic and foreign production (PQ), household demand (HHX), labor demand (LAB), capital demand (CAP), total production (Q), import (QM), total consumption (X), total export (XE), domestic production (QD), household income (Inc), and price level (PL). This report specifically focuses on the impact of energy shocks stemming from positive labor and capital productivity since recent research suggests that it is enhanced energy productivity rather than a direct increase in supply that is the key to meeting the challenges posed by energy problems¹.

Literature Review

The effects of energy shocks on economic variables are not easily discernible. Recent observations, according to Kilian (2008), show that global political dynamics and patterns of demand and supply have a greater impact on price fluctuations after an energy shock than changes in factor productivity within the domestic marketⁱⁱ. However, in this particular economy, the present paper aims to examine the effects of a marginal productivity shock through the utilization of a highly simplified economic model. The initial impact of a positive productivity shock will lead to a rise in energy production and a decrease in production costs due to economies of scale. However, the extent of the enhanced output will mostly be contingent upon the specific shape of the aggregate demand curveⁱⁱⁱ. This would result in a notable augmentation of energy provision for companies that bear a substantial proportion of energy expenses, such as petroleum refining and industrial chemicals. Additionally, it will generate a surge in demand within other pertinent sectors, such as the automobile industry^{iv}. This fall in energy prices will also translate into real income gains in the economy, which in turn will positively influence the demand for energy as well as numerous categories of durable goods^v.

For example, energy consumption in houses is mostly allocated to transportation and heating purposes. Consequently, their energy requirement encompasses a need for services that necessitate the utilization of long-lasting commodities. Nevertheless, when various durable products belonging to the same category are paired with varying quantities of energy, they produce identical services. Therefore, a reduction in energy prices would lead to an increase in the utilization of durable items already in circulation, thereby facilitating more energy consumption. However, as mentioned previously, as a result of the reduced price, there would also be an increase in real income. Consequently, the decrease in energy prices would impact private consumption through both direct and indirect means. Additionally, the changes in energy prices will result in a corresponding adjustment in the combination of productive resources utilized by enterprises. This could potentially result in heightened investment and a subsequent augmentation of capital accumulation in the future.

The impact of reduced energy prices will be more pronounced on the economies of energy-exporting regions, such as Wyoming, compared to energy-importing regions, leading to a rise in output^{vi}. The reason behind this phenomenon is that places that have a surplus of energy and export it to other regions tend to experience a positive impact on their terms of trade when energy prices decrease^{vii}.

Model & Methodology

This study collected all quantitative data from the SAM (social accounting matrix) table pertaining to trade and product differentiation. This approach was chosen to ensure that the data accurately represented the cyclical flow of products, services, and cash within an economy. The model exhibits conditions of zero profit, market clearance, and income balance. Once the table has been formulated, the benchmark equilibrium is established. Subsequently, benchmark data is utilized to parameterize the behavioral model through a calibration process. Subsequently, the model is executed utilizing the calibrated parameters in order to verify its ability to reproduce the benchmark equilibrium. After addressing the aforementioned matter, a policy was implemented within the GAMs model, namely in the form of a positive productivity shock for both capital and labor in the energy industry. The resulting percentage changes from the benchmark level in the dependent variables were subsequently observed for the purpose of policy analysis.

In this model, a constant elasticity of substitution (CES) function was used within the model due to its ability to accommodate less restrictive assumptions regarding the interplay between capital and labor in the production process. In this context, the parameter θ represents the benchmark value share of each industry. The elasticity of substitution, denoted by σ , characterizes the degree to which consumers can interchangeably consume different combinations of goods. The substitution parameter, ρ , is defined as $(\sigma - 1)/\sigma$. Additionally, the parameter α is associated with the expenditure allocated to a particular good within the utility function.

At market equilibrium,

- (i) Price of total consumption * total consumption = price in domestic market * household demand + export price * export.
- (ii) Price of total production * total production = price in domestic market * domestic production + import price * import.
- (iii) Price in domestic market * domestic production = wage rate * labor + rental rate * capital (at zero profit) = income.

Based on the data presented in the SAM table representing a general equilibrium, it is evident that among the four industries under consideration, the energy sector exhibits the highest level of capital intensity, utilizing capital resources at a ratio of 1.7 times that of labor. Subsequently, the agriculture industry follows suit with a capital-to-labor ratio of 1.4. The sectors of recreation and miscellaneous exhibit a higher degree of labor intensity, requiring approximately 1.1 times more labor input compared to capital input. In the context of trade, it is seen that the majority of these items are traded rather than being consumed domestically. Notably, the energy sector stands out as the largest contributor in this module, with a significant disparity of 230 times between trade volume and domestic demand. Agriculture exhibits a significantly higher level of trade compared to household demand, with a ratio of 5.2. In contrast, entertainment and miscellaneous sectors jointly occupy the third position, with trade surpassing domestic demand by a factor of approximately 1.1. According to the configuration of the SAM table, it is anticipated that an energy shock should yield a substantial effect on net exports (if none of the relevant variables were held constant). Furthermore, it is important to highlight that the labor-to-capital intensity and export sensitivity of recreation and miscellaneous goods exhibit uniformity throughout the model. Consequently, any policy consequence should ideally produce equivalent effects on both sectors in terms of both direction and magnitude. In order to assess the impact of a policy application on welfare,

the compensating variation (CV) is employed as a measure. The CV quantifies whether consumers would require compensation to maintain the same level of utility as before (indicating a welfare loss) or if they would need to make payments to achieve the same level of well-being as previously (indicating a welfare gain).

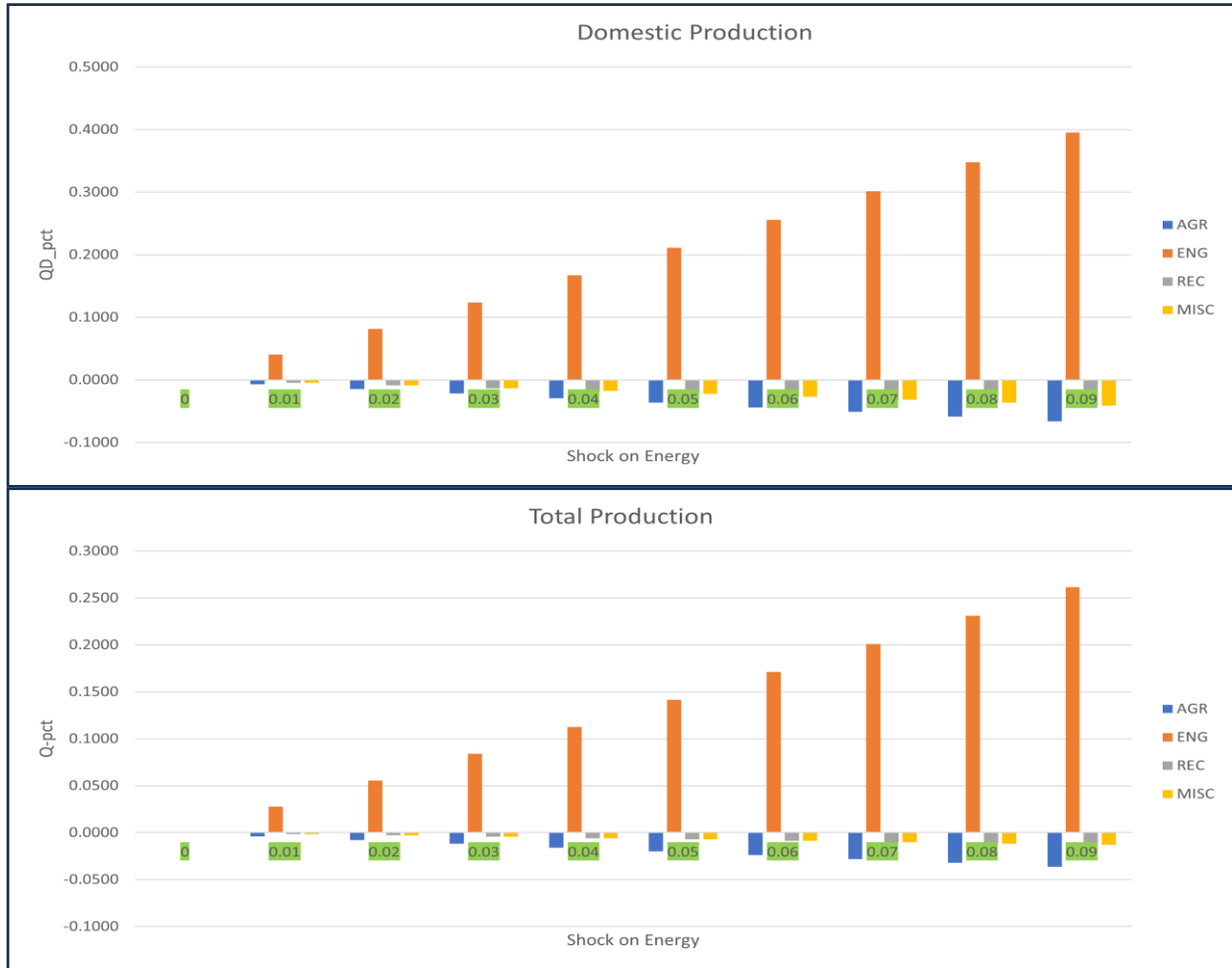
In the context of GAMs (General Algebraic Modeling System), the benchmark values for the target variables were established, and the import and export prices (PMQ and PEX) were set at a constant level. So, there is no change in foreign exchange rates. Furthermore, it is important to note that the model does not incorporate a governmental entity. As a result, all benefits and losses are solely dispersed through households, enterprises, and in the context of trade. The shock parameter was initially introduced in the GAMs model in the form of $shkk(i,tt)=shk(i)*ord(tt)-1$; and subsequently, a positive shock was incrementally incorporated into the factor demands. The factor demands after the shock had the following equations:

$$1. LAB_p_eq(i,tt) \dots LAB_p(i,tt)=e=(QD_p(i,tt)/(theta(i)*(1+shkk(i,tt))))*(((1-\beta(i))/w_p(tt))^{**\sigma(i)})*(((\beta(i))^{**\sigma(i)})*(r_p(tt)^{(1-\sigma(i))})+((1-\beta(i))^{**\sigma(i)})*(w_p(tt)^{(1-\sigma(i))}))^{**(\sigma(i)/(1-\sigma(i))));$$

$$2. CAP_p_eq(i,tt) \dots CAP_p(i,tt)=e=(QD_p(i,tt)/(theta(i)*(1+shkk(i,tt))))*(((\beta(i))/r_p(tt))^{**\sigma(i)})*(((\beta(i))^{**\sigma(i)})*(r_p(tt)^{(1-\sigma(i))})+((1-\beta(i))^{**\sigma(i)})*(w_p(tt)^{(1-\sigma(i))}))^{**(\sigma(i)/(1-\sigma(i))));$$

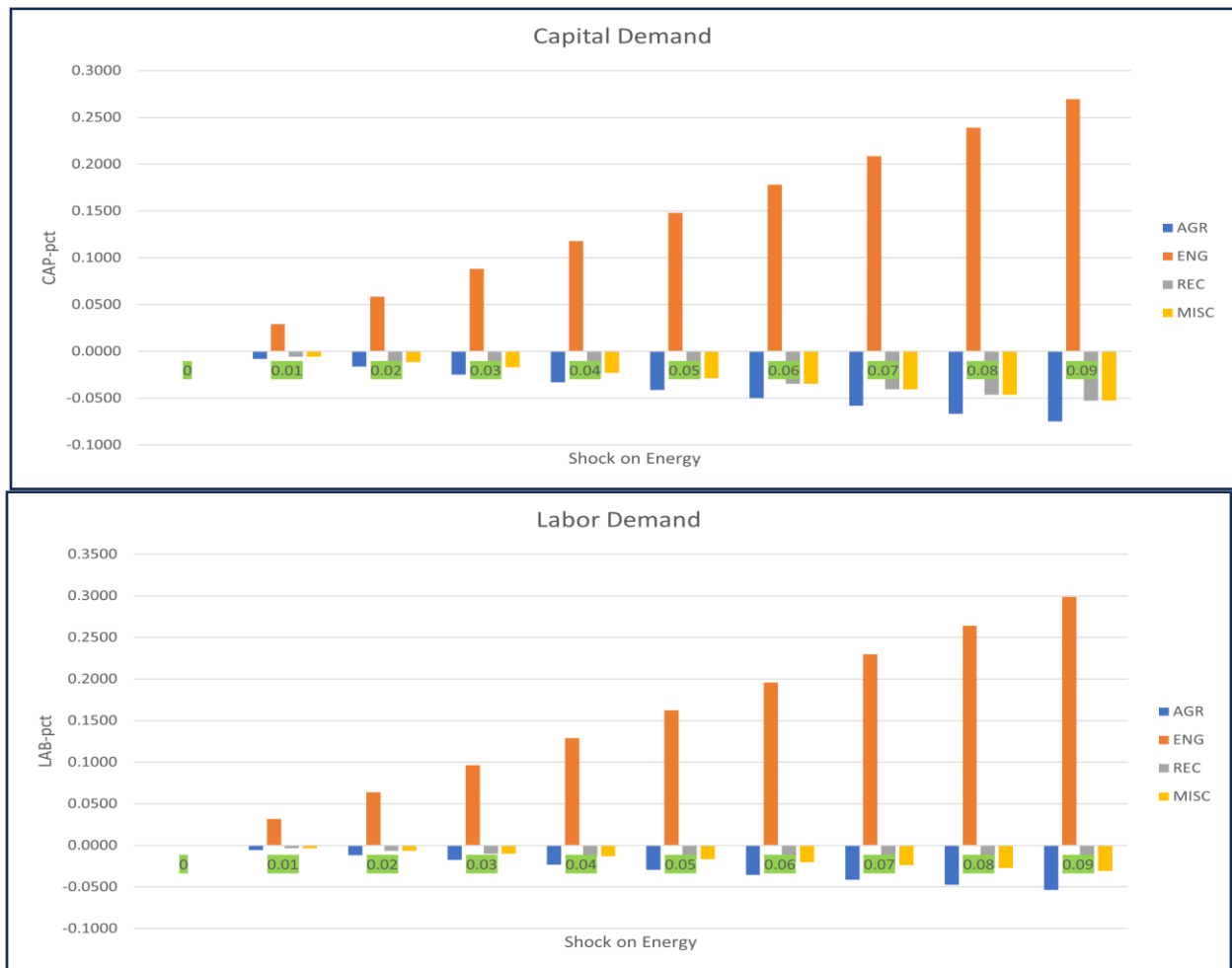
The subsequent section will illustrate and analyze the alterations in the 16 dependent variables resulting from the implementation of the shock.

Results and Discussion

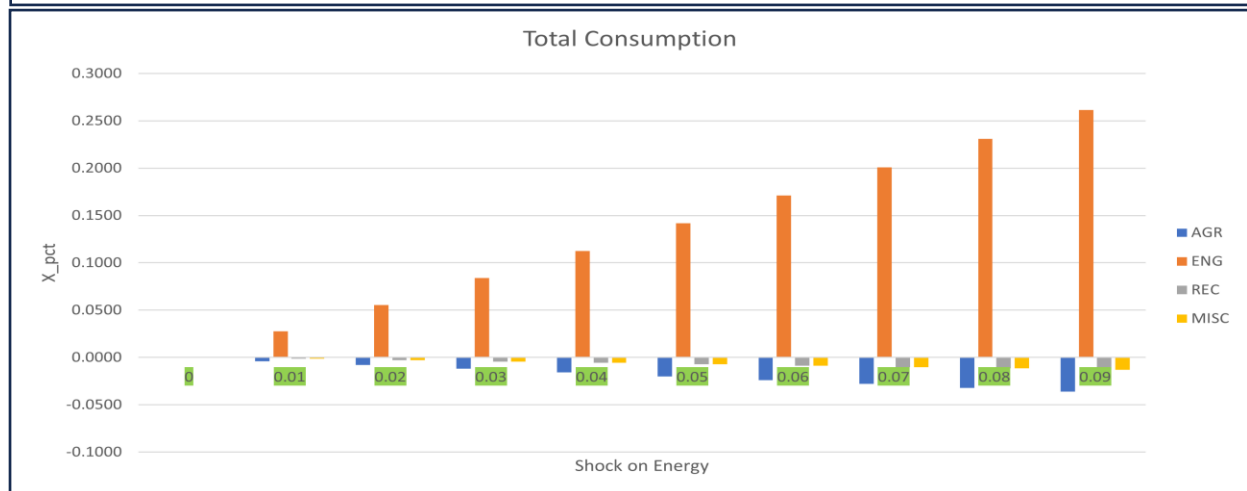
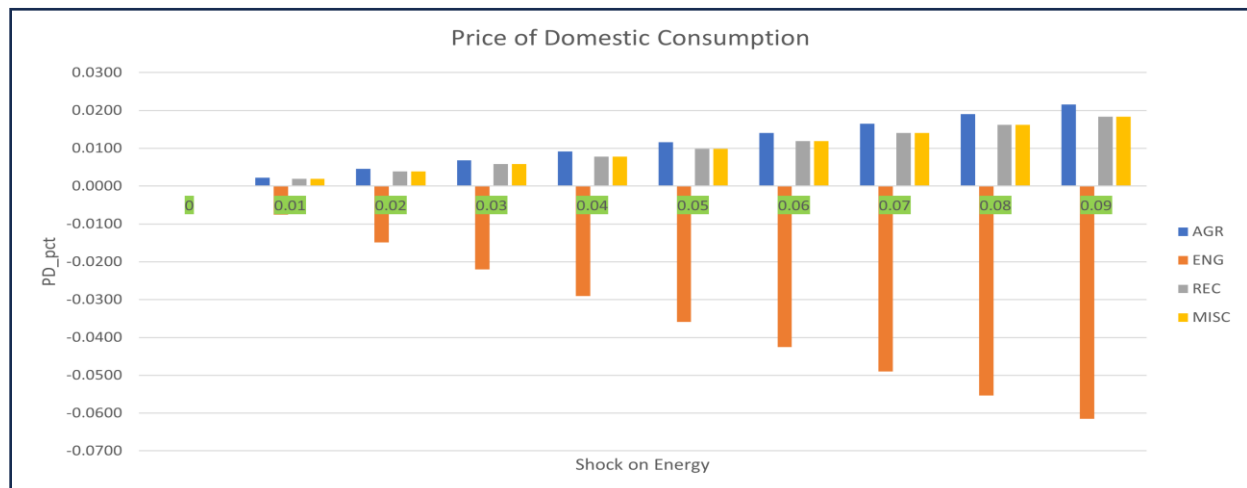
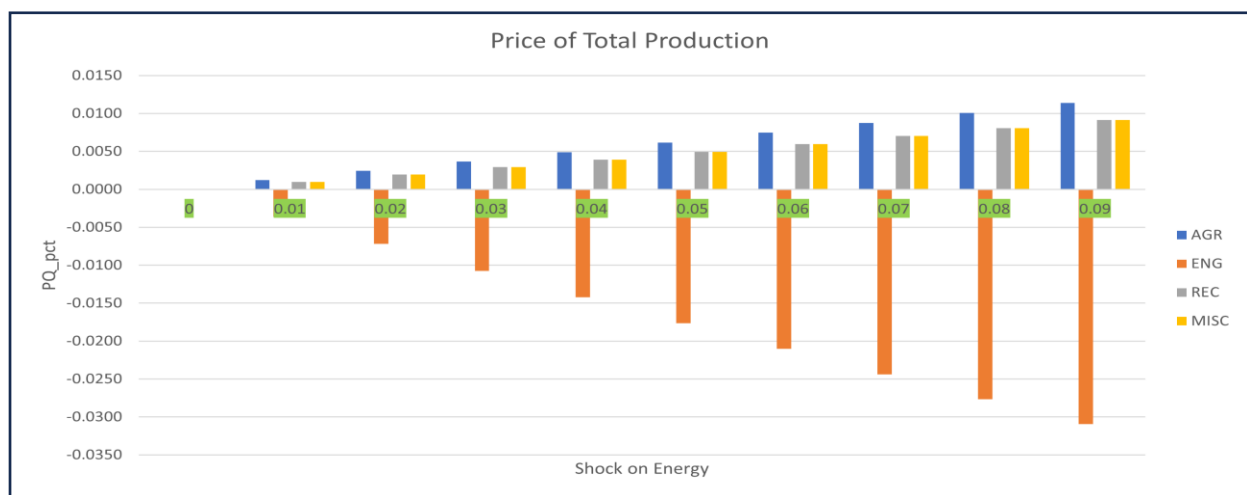


The immediate consequence of the positive energy shock, in relation to the augmentation of factor productivity, would be the increase of domestic production (QD) and therefore the total production (Q) within the energy sector. In the event of a positive shock ranging from 1% to 9% in the energy sector, the domestic production for energy is projected to experience an increase of about 4.0% to 39.5%. Similarly, the total output is expected to rise by approximately 2.8% to 26.1%. However, to facilitate the increased energy production, the factors of production would be transferred from the other industries to the energy sector, which would result in a decrease in their respective outputs. Given the capital-intensive nature of both energy and agriculture sectors, it is expected that a shock of 9% would result in a higher decline in domestic and total agricultural production (up to 6.6% and 3.6%, respectively) compared to the decline in recreation and miscellaneous sectors (up to 4.1% and 1.3%, respectively). It is noteworthy to acknowledge that the decline in agricultural output, while notable, would not be as substantial as the rise in energy production. This discrepancy may be attributed to two factors: firstly, the goods in question are not perfect

substitutes, and secondly, the agricultural sector is not as reliant on capital-intensive processes as the energy sector.



With the aforementioned factor transfer to energy from other industries, the factor demands, both capital (CAP) and labor (LAB), would increase proportionally to the increase in production in the energy industry and decrease accordingly for the other 3 goods.



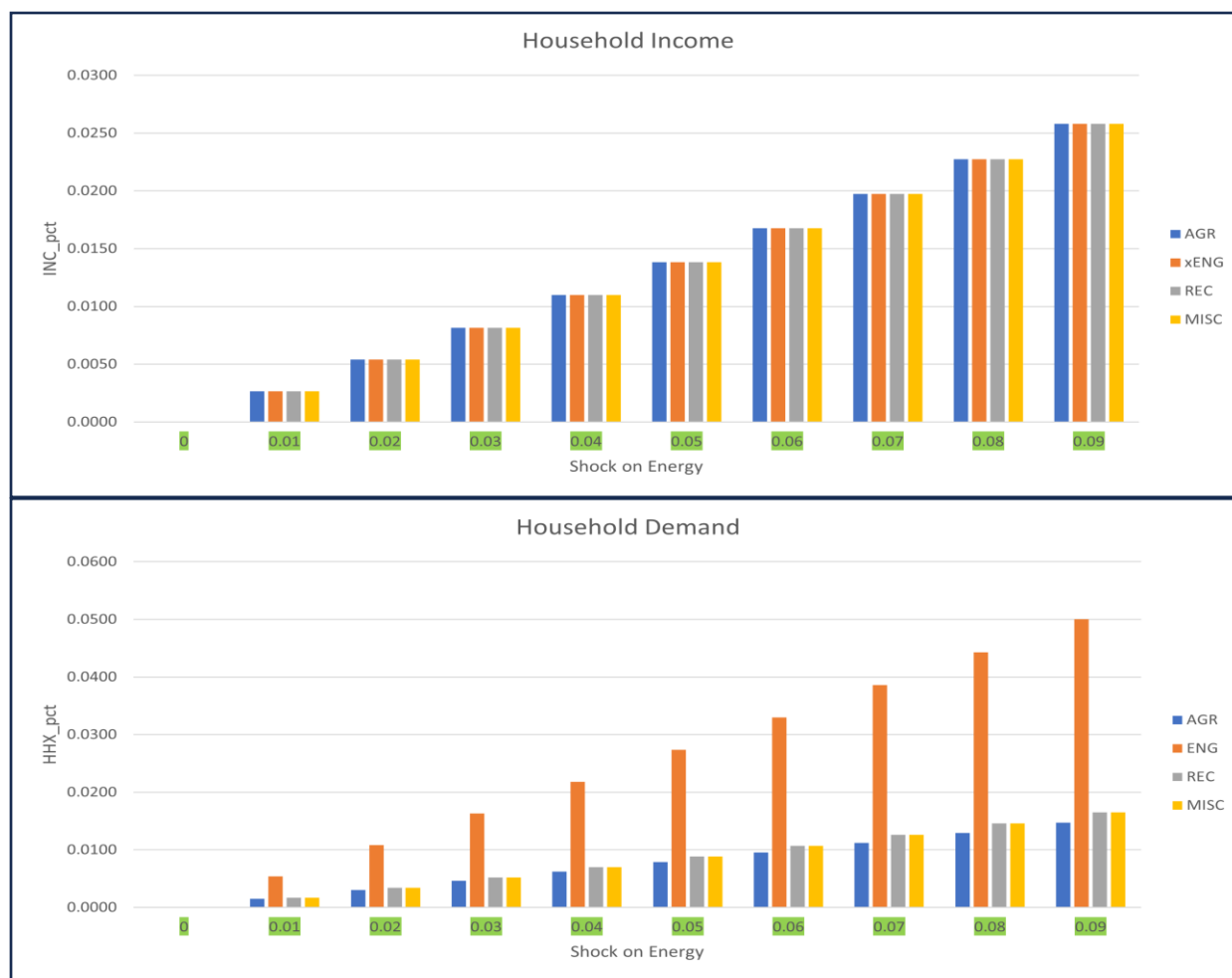
Another proportional change would occur in the prices of total production (PQ), as increased energy production would lead to a corresponding decrease in energy production prices due to economies of scale. Simultaneously, the production prices for other goods would rise by the same proportion as the reduction in production.

The observed rise in energy production and corresponding decline in production across other sectors would result in an augmented energy supply and a reduced supply of agricultural,

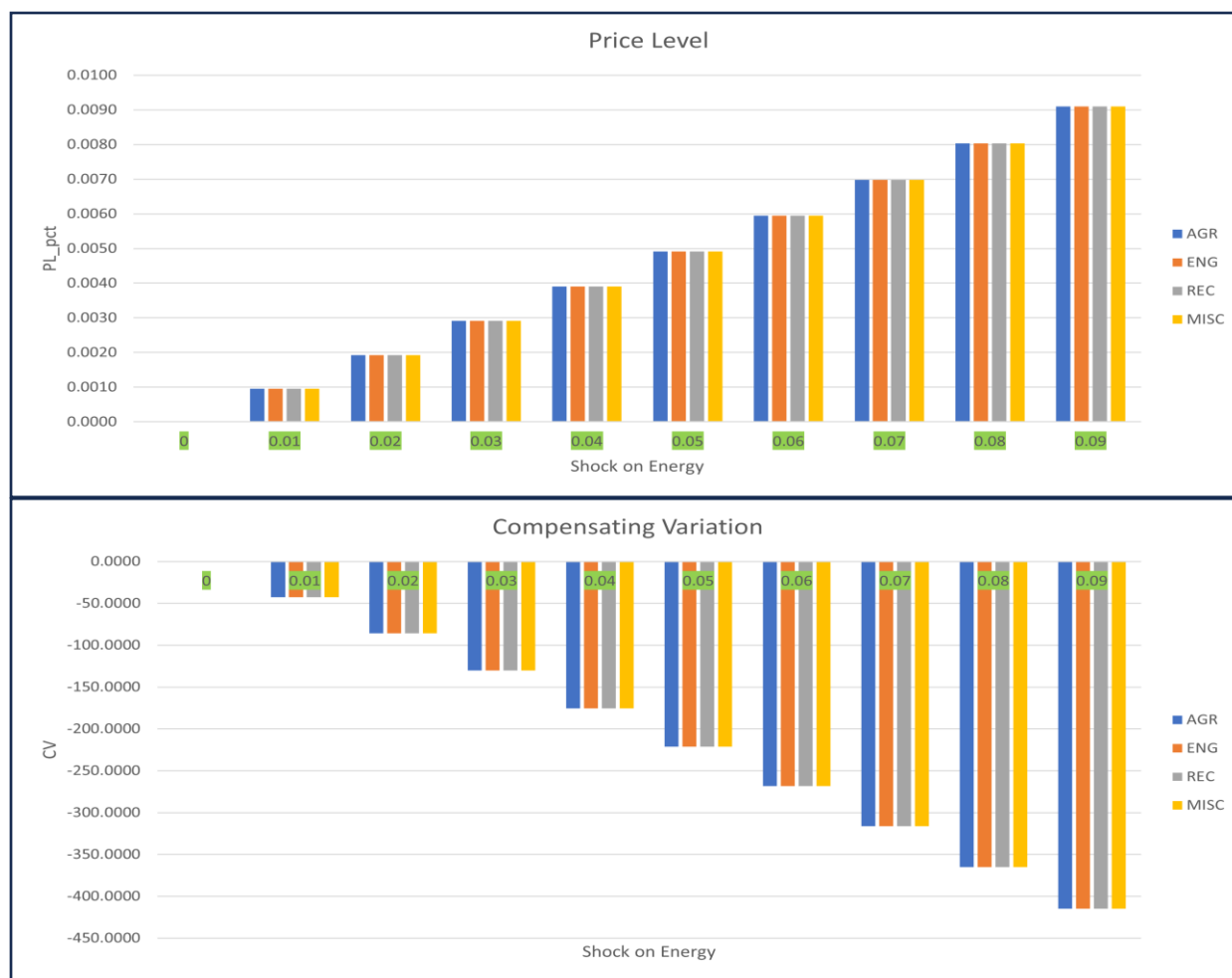
recreational, and miscellaneous commodities. This increased and decreased supply of energy and other goods would then inflict downward and upward pressure on those goods, respectively, causing the price of domestic consumption (PD) to go down and up accordingly for energy and the rest of the goods. However, since consumption is price-sensitive (inversely related), with lower prices in energy, total consumption (X) of energy would go up parallelly, while for the same reason, total consumption of the other goods would fall in an equivalent manner.



In the context of the factor market, an increase in the demand for labor and capital in the energy sector would lead to a corresponding rise in the prices of labor (wage rate, denoted as w) and capital (rental rate, denoted as r). However, as wages and rent in the energy sector continue to rise, there is a likelihood of labor and capital being reallocated from other sectors to the energy sector. This could potentially lead to instability in the factor market until a new equilibrium is reached. This equilibrium would be characterized by symmetric increases in wages and rent across all industries.



Given that wages and rents serve as the primary sources of income for households in practice, it follows that an increase in wages and rents would result in a corresponding increase in household income (INC). In the realm of the product market, the rise in income exerts upward pressure on household demand (HHX) for all commodities. However, it is important to note that this surge in demand is not uniform across various industries. The increase in demand for energy will significantly surpass the growth observed in other industries. This phenomenon can be attributed to the concurrent increase in wealth and decrease in energy prices, a trend that is not observed in other industries. However, it is important to note that there also exists a clear divergence in the rate of HHX growth between the agricultural sector and the other two industries. Given the higher percentage increase in prices for agricultural goods, attributed to their capital-intensive nature (due to capital competition to energy), it can be inferred that the rise in household demand for agricultural products would be comparatively lower than that for recreation and miscellaneous commodities. This can be attributed to the price sensitivity exhibited by consumers.



The rise in income and subsequent increase in household demand would then lead consumers to engage in higher levels of expenditure, resulting in a rightward shift of the aggregate demand curve in the product market. Consequently, the price level (PL) would experience an upward trajectory within the economy, leading to inflationary pressures in the short term. However, it is important to note that the rise in the price level is quite modest when compared to the growth in income. Consequently, consumers in this society would ultimately experience an improvement in their overall wellbeing. The welfare gain can be demonstrated through the decline in compensatory variation (CV). In other words, individuals would need to relinquish their earnings in order to maintain the same degree of utility as before.



The potential increase in energy production and decrease in energy prices would most likely lead to a significant rise in overseas demand for energy items. Consequently, the economy would likely engage in a substantial amount of exporting (XE) for energy in response. However, due to the significant surplus in energy exports, the region would also be compelled to import (QM) various

types of energy commodities in order to satisfy the growing demand from local households. Moreover, with the positive energy shock and subsequent significant rise in output, exports, and decreased domestic prices, it is reasonable to anticipate a corresponding decrease in the prices of overall consumption (PX), both domestic and foreign, as a result of the shock's impact.

For recreation and miscellaneous goods, with domestic prices increasing, exports would fall and imports would rise accordingly. But since production falls and household demand increases, import prices would also catch up soon, and the prices of total consumption would go up by a visible margin.

In the agriculture industry, the magnitude of reduction in production is quite sizable (since it's capital-intensive like energy and so it will suffer more as the positive shock in energy takes away more capital from it), and so is the rise in the prices of domestic consumption and production. Consequently, there is a notable increase in both domestic consumption and production prices. This phenomenon would then result in a significant decline in exports, as comparative advantage diminishes, and an increase in overall consumption prices for agricultural products. Nevertheless, it is somewhat unusual that an incremental shock results in a marginal decrease in agricultural imports. This counter-intuitive outcome contradicts the expectation that increased household demand and domestic prices, together with reduced output and exports, would lead to a greater need for agricultural imports. The observed modification may arise from the unaccounted delays in the bookkeeping process of the SAM model, wherein a sum of \$8753.55 is included in the trade and household demand cell. However, given the minute nature of this alteration, it is plausible that other systematic errors or external variables could also account for this occurrence.

Conclusion

This report aimed to investigate the effects of a policy shift, namely a positive supply shock characterized by an increase in factor productivity in the energy sector, on 16 pertinent economic variables across four industries. The experimental data is derived from a normalized Social Accounting Matrix (SAM) model, wherein all units are converted into monetary terms (expressed in dollars). Consequently, in the benchmark scenario, all prices are precisely equal to 1. Once the data was imported into GAMs and the model's validity was assessed, a policy was implemented to examine the percentage changes in all 16 variables across various industries. Subsequently, these changes were evaluated in relation to economic intuition (hypothesized), and the underlying interactive factors among the variables were identified. The analysis revealed that a productivity shock in the energy sector would have varying effects on the variables associated with all four goods, and none of the variables would remain completely unaffected, highlighting the interconnectedness among these variables.

It has been found that different industries exhibit different responses to the shock, which can be attributed to their varying levels of factor intensity and trade-to-domestic demand ratio. According to the SAM model, it is seen that the factor and trade intensity of recreation and miscellaneous goods and services were nearly identical (in terms of ratio). Consequently, all the resultant changes in variables resulting from the policy shock were analogous for these two categories of commodities. On the other hand, there was a mentionable relationship between the changes in agricultural and energy products due to their shared preference for capital compared to labor inputs.

The alterations in the dependent variables largely align with the anticipated adjustments that were documented in the prior research studies cited in the literature review section. There is significant potential for further investigation with this particular form of analysis. For instance, this model could be expanded to facilitate more economic variables and sectors (like government) to analyze the interactions of more economic parameters and better understand the nature of the economy when a relevant policy is introduced.

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Appendix

step	industry	shkk	CV	PD_pct	w_pct	r_pct	PX_pct	PQ_pct	HHX_pct	LAB_pct	CAP_pct	Q_pct	QM_pct	X_pct	XE_pct	QD_pct	INC_pct	PL_pct
1	AGR	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	ENG	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	REC	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	MISC	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	AGR	0.0000	-42.5195	0.0022	0.0005	0.0035	0.0004	0.0012	0.0015	-0.0058	-0.0082	-0.0039	-0.0001	-0.0039	-0.0050	-0.0072	0.0027	0.0010
2	ENG	0.0100	-42.5195	-0.0075	0.0005	0.0035	0.0000	-0.0036	0.0054	0.0317	0.0292	0.0276	0.0159	0.0276	0.0277	0.0404	0.0027	0.0010
2	REC	0.0000	-42.5195	0.0019	0.0005	0.0035	0.0008	0.0010	0.0017	-0.0032	-0.0057	-0.0014	0.0016	-0.0014	-0.0038	-0.0044	0.0027	0.0010
2	MISC	0.0000	-42.5195	0.0019	0.0005	0.0035	0.0008	0.0010	0.0017	-0.0032	-0.0057	-0.0014	0.0016	-0.0014	-0.0038	-0.0044	0.0027	0.0010
3	AGR	0.0000	-85.9245	0.0045	0.0009	0.0071	0.0007	0.0024	0.0031	-0.0116	-0.0164	-0.0079	-0.0003	-0.0079	-0.0100	-0.0144	0.0054	0.0019
3	ENG	0.0200	-85.9245	-0.0149	0.0009	0.0071	-0.0001	-0.0072	0.0108	0.0638	0.0586	0.0556	0.0318	0.0556	0.0558	0.0817	0.0054	0.0019
3	REC	0.0000	-85.9245	0.0038	0.0009	0.0071	0.0017	0.0019	0.0034	-0.0065	-0.0114	-0.0029	0.0032	-0.0029	-0.0077	-0.0088	0.0054	0.0019
3	MISC	0.0000	-85.9245	0.0038	0.0009	0.0071	0.0017	0.0019	0.0034	-0.0065	-0.0114	-0.0029	0.0032	-0.0029	-0.0077	-0.0088	0.0054	0.0019
4	AGR	0.0000	-130.2166	0.0069	0.0014	0.0107	0.0011	0.0037	0.0046	-0.0175	-0.0247	-0.0119	-0.0004	-0.0119	-0.0151	-0.0217	0.0082	0.0029
4	ENG	0.0300	-130.2166	-0.0220	0.0014	0.0107	-0.0001	-0.0107	0.0163	0.0963	0.0882	0.0839	0.0477	0.0839	0.0842	0.1239	0.0082	0.0029
4	REC	0.0000	-130.2166	0.0058	0.0014	0.0107	0.0025	0.0029	0.0052	-0.0099	-0.0171	-0.0043	0.0049	-0.0043	-0.0116	-0.0133	0.0082	0.0029
4	MISC	0.0000	-130.2166	0.0058	0.0014	0.0107	0.0025	0.0029	0.0052	-0.0099	-0.0171	-0.0043	0.0049	-0.0043	-0.0116	-0.0133	0.0082	0.0029
5	AGR	0.0000	-175.3971	0.0092	0.0020	0.0144	0.0015	0.0049	0.0062	-0.0234	-0.0330	-0.0159	-0.0006	-0.0159	-0.0202	-0.0290	0.0110	0.0039
5	ENG	0.0400	-175.3971	-0.0290	0.0020	0.0144	-0.0001	-0.0142	0.0218	0.1291	0.1180	0.1126	0.0635	0.1126	0.1130	0.1670	0.0110	0.0039
5	REC	0.0000	-175.3971	0.0078	0.0020	0.0144	0.0034	0.0039	0.0070	-0.0132	-0.0229	-0.0058	0.0066	-0.0058	-0.0155	-0.0178	0.0110	0.0039
5	MISC	0.0000	-175.3971	0.0078	0.0020	0.0144	0.0034	0.0039	0.0070	-0.0132	-0.0229	-0.0058	0.0066	-0.0058	-0.0155	-0.0178	0.0110	0.0039
6	AGR	0.0000	-221.4671	0.0116	0.0025	0.0182	0.0019	0.0062	0.0079	-0.0294	-0.0414	-0.0199	-0.0007	-0.0199	-0.0253	-0.0364	0.0138	0.0049
6	ENG	0.0500	-221.4671	-0.0359	0.0025	0.0182	-0.0001	-0.0176	0.0274	0.1624	0.1480	0.1416	0.0794	0.1416	0.1421	0.2110	0.0138	0.0049
6	REC	0.0000	-221.4671	0.0099	0.0025	0.0182	0.0043	0.0049	0.0089	-0.0166	-0.0288	-0.0072	0.0083	-0.0072	-0.0195	-0.0224	0.0138	0.0049
6	MISC	0.0000	-221.4671	0.0099	0.0025	0.0182	0.0043	0.0049	0.0089	-0.0166	-0.0288	-0.0072	0.0083	-0.0072	-0.0195	-0.0224	0.0138	0.0049
7	AGR	0.0000	-268.4275	0.0141	0.0030	0.0220	0.0023	0.0075	0.0095	-0.0354	-0.0497	-0.0240	-0.0008	-0.0240	-0.0305	-0.0438	0.0168	0.0059
7	ENG	0.0600	-268.4275	-0.0425	0.0030	0.0220	-0.0002	-0.0210	0.0330	0.1959	0.1782	0.1711	0.0952	0.1711	0.1716	0.2558	0.0168	0.0059
7	REC	0.0000	-268.4275	0.0120	0.0030	0.0220	0.0052	0.0060	0.0107	-0.0201	-0.0346	-0.0087	0.0101	-0.0087	-0.0235	-0.0270	0.0168	0.0059
7	MISC	0.0000	-268.4275	0.0120	0.0030	0.0220	0.0052	0.0060	0.0107	-0.0201	-0.0346	-0.0087	0.0101	-0.0087	-0.0235	-0.0270	0.0168	0.0059
8	AGR	0.0000	-316.2790	0.0165	0.0036	0.0258	0.0027	0.0088	0.0112	-0.0415	-0.0581	-0.0281	-0.0010	-0.0281	-0.0357	-0.0513	0.0197	0.0070
8	ENG	0.0700	-316.2790	-0.0490	0.0036	0.0258	-0.0002	-0.0244	0.0386	0.2299	0.2085	0.2008	0.1110	0.2008	0.2015	0.3015	0.0197	0.0070
8	REC	0.0000	-316.2790	0.0141	0.0036	0.0258	0.0061	0.0070	0.0126	-0.0236	-0.0405	-0.0102	0.0119	-0.0102	-0.0275	-0.0316	0.0197	0.0070
8	MISC	0.0000	-316.2790	0.0141	0.0036	0.0258	0.0061	0.0070	0.0126	-0.0236	-0.0405	-0.0102	0.0119	-0.0102	-0.0275	-0.0316	0.0197	0.0070
9	AGR	0.0000	-365.0222	0.0191	0.0042	0.0297	0.0032	0.0101	0.0130	-0.0476	-0.0666	-0.0322	-0.0011	-0.0322	-0.0410	-0.0587	0.0227	0.0080
9	ENG	0.0800	-365.0222	-0.0553	0.0042	0.0297	-0.0002	-0.0277	0.0443	0.2641	0.2390	0.2309	0.1268	0.2309	0.2317	0.3480	0.0227	0.0080
9	REC	0.0000	-365.0222	0.0162	0.0042	0.0297	0.0071	0.0081	0.0146	-0.0271	-0.0465	-0.0116	0.0137	-0.0116	-0.0316	-0.0363	0.0227	0.0080
9	MISC	0.0000	-365.0222	0.0162	0.0042	0.0297	0.0071	0.0081	0.0146	-0.0271	-0.0465	-0.0116	0.0137	-0.0116	-0.0316	-0.0363	0.0227	0.0080
10	AGR	0.0000	-414.6574	0.0216	0.0048	0.0337	0.0036	0.0114	0.0147	-0.0537	-0.0750	-0.0363	-0.0012	-0.0363	-0.0463	-0.0662	0.0258	0.0091
10	ENG	0.0900	-414.6574	-0.0615	0.0048	0.0337	-0.0002	-0.0309	0.0500	0.2987	0.2695	0.2614	0.1425	0.2614	0.2623	0.3954	0.0258	0.0091
10	REC	0.0000	-414.6574	0.0184	0.0048	0.0337	0.0080	0.0091	0.0165	-0.0306	-0.0524	-0.0131	0.0156	-0.0131	-0.0358	-0.0410	0.0258	0.0091
10	MISC	0.0000	-414.6574	0.0184	0.0048	0.0337	0.0080	0.0091	0.0165	-0.0306	-0.0524	-0.0131	0.0156	-0.0131	-0.0358	-0.0410	0.0258	0.0091